

● RS002 Week 12 — English Worksheet

Topic: Refactor the Whole App into Functions · **Course:** Pokédex App · **Time:** about 45 minutes

This week you tidy everything. All those lines you wrote one by one over the last 11 weeks get grouped into **named functions**, and your main code shrinks to a short **menu loop** that just *calls* them. This is called **refactoring** — same behaviour, cleaner shape.

1 · Recall

From memory:

Write a function `count_dex(pokedex)` **that returns how many Pokémon are in the list.**

2 · Predict 🌟

Read each block. Before you run it, write what you think will happen.

```
def show_menu():
    print("1. 도감 보기")
    print("2. 종료")

show_menu()
choice = input("선택: ").strip()
print(f"고른 메뉴: {choice}")
```

The user types **1**. What is printed in total?


```
while True:
    cmd = input("명령 (q로 종료): ").strip()
    if cmd == "q":
        break
    print(f"입력: {cmd}")
```

The user types **a**, then **b**, then **q**. What does each round do, and when does it stop?


```
trainer = input("이름: ").strip() or "이름 모를 트레이너"
print(trainer)
```

The user presses Enter without typing. What prints? What does `or` do here?

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it, then explain the fix.

Bug A — A function must be **defined above** where it is called. Right now `show_menu` is called before it exists, so the program crashes.

```
show_menu()

def show_menu():
    print("=== 메뉴 ===")
```

Hint: move the definition above the call.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This menu loop should keep running until the user picks `3`. Right now it runs once and stops.

```
choice = input("선택: ").strip()
if choice == "1":
    print("도감 보기")
elif choice == "3":
    print("종료")
```

Hint: wrap it in `while True:` and `break` on `3`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — When the user types something that is not a menu number, the app should say so. Right now an unknown choice silently does nothing.

```
while True:
    choice = input("선택: ").strip()
    if choice == "1":
        print("도감 보기")
    elif choice == "2":
        break
```

Hint: add an `else` for unknown input.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Start from this working function-based app:

```

def show_intro(trainer_name):
    print(f"\n=== 포켓몬 도감 ===")
    print(f"환영합니다, {trainer_name} 트레이너님!\n")

def show_menu():
    print("\n--- 메뉴 ---")
    print("1. 도감 보기")
    print("2. 포켓몬 검색")
    print("3. 종료")

def list_all(pokedex):
    for p in pokedex:
        print(f"  {p['name']} 타입:{p['type']} HP:{p['hp']}")

def search_pokemon(pokedex):
    keyword = input("검색어: ").strip().lower()
    matches = [p for p in pokedex if keyword in p["name"].lower()]
    if matches:
        for p in matches:
            print(f"  {p['name']} ({p['type']})")
    else:
        print("검색 결과 없음")

pokedex = [
    {"name": "피카츄", "type": "전기", "hp": 35},
    {"name": "이상해씨", "type": "풀", "hp": 45},
]

trainer = input("트레이너 이름: ").strip() or "이름 모를 트레이너"
show_intro(trainer)

while True:
    show_menu()
    choice = input("선택: ").strip()
    if choice == "1":
        list_all(pokedex)
    elif choice == "2":
        search_pokemon(pokedex)
    elif choice == "3":
        print("또 만나요!")
        break
    else:
        print("1~3 중에서 골라주세요.")

```

Make these changes one at a time and run after each:

1. Add a new menu option `4. 포켓몬 수 보기` that prints `len(pokedex)` .
2. Move the dex count into its own function and call it from the menu.
3. Make the intro show how many Pokémon are registered.

Write your changed / added lines here:

5 · Make It 📷

Refactor your **whole Week 11 app** so the main code is a short **menu loop** that only calls functions. Aim for a main section under ~10 lines. Every action (list, search, show) lives in its own function.

When it works, send a **video** on KakaoTalk and tell us **how many functions** you made. Then explain what you did. Use these starters — write 4 to 6 sentences.

First, I grouped my code into functions called ...

My main loop now only ...

I made ... functions in total.

Functions must be defined before they are called because ...

The trickiest part was ...

Refactoring did not change what the app does, but it ...

6 · Record Your Walkthrough 🎥

Film it running. Teach it like the viewer is new. Try to use: **refactor**, **function**, **menu loop**, **call**, **break**.

1. Run the menu and try every option, including quit.
2. Scroll to your main loop and show how short it is.
3. Point to one function and explain what the loop *calls*.
4. Say why a short main loop is easier to read and fix.

Plan what you will say here first:

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.